Chapter 7: Shapes and Patterns

Worksheet 7: Shapes and Patterns

Learning Objective: Identify and classify 2D and 3D shapes by their properties; recognise, extend and create patterns; calculate the perimeter of simple shapes.

Worked Example:
Find the perimeter of a rectangular garden that is 15 m long and 9 m wide.

Solution: Perimeter $= 2(l + w) = 2(15 + 9) = 2 \times 24 = 48$ m.

Questions (1–20)

1. How many sides does a hexagon have?

2. Name a 3D shape that has no flat faces.

3. How many faces does a cube have?

4. What is the next shape in this pattern: circle, square, circle, square, _____?

5. How many corners (vertices) does a triangle have?

6. A cone has how many curved surfaces and how many flat faces?

7. Find the perimeter of a square park with side 25 m.

8. A triangular flag has sides 12 cm, 12 cm and 8 cm. Find its perimeter.

9. Continue the number pattern: 2, 4, 6, 8, _____, _____.

10. A cuboid-shaped box has length 10 cm, breadth 5 cm and height 4 cm. How many faces does it have?

11. Find the perimeter of a rectangular field 80 m long and 50 m wide. If a farmer fences it, how much wire is needed?

12. A tile pattern repeats: Red, Blue, Blue, Red, Blue, Blue, . . . What colour is the 10th tile?

13. Continue the pattern: 1, 3, 5, 7, _____, _____.

14. A pentagon has equal sides of 9 cm each. Find its perimeter.

15. How many edges does a cube have?

16. A rectangular plot has a perimeter of 60 m. If its length is 18 m, find its width.

17. A pattern repeats every 4 shapes: square, circle, triangle, triangle. What will be the 23rd shape?

18. Two square tiles of side 5 cm are joined edge-to-edge to form a rectangle. Find the perimeter of the new rectangle.

19. Look at the floor tiles, a saree border, or a rangoli design at home. Describe the repeating pattern you see.

20. Measure the length and width of your classroom door or a window, and calculate its perimeter.

Reflection Question: Find one object at home shaped like a 3D shape you learned about (cube, cone, cylinder, sphere). Name the object and the shape.

Real-Life Application Question: Look around your house for a pattern (on clothes, walls, or floor tiles). Describe it and say what comes next in the pattern.